

RAY: Resilience Assistant for You

An Accountable GeoAI System for Place-Based Disaster Intelligence

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Abstract

Vulnerability, risk and priority are value-laden; a system that computes them for the public owes the public an account of *who* decided *what*, on *which* evidence, at *what* resolution. RAY (Resilience Assistant for You) is a disaster-resilience agent built around that obligation. Its pipelines run autonomously — hourly federal hazard feeds, and deep-case builders that turn raw damage registries, social vulnerability, census population and street imagery into H3 grids across three events in two hazard types — but every stage runs inside the *Steward Harness*, which enforces outcome, process and institutional validity in code: spatial checks and mandatory uncertainty per tile, SHA-256 lineage in an append-only audit, a declarative policy of who may be told what at which resolution, a distribution policy verified by a CI gate, and citation-by-default for every sentence the agent utters. The value judgements are made explicit and handed back to people: vulnerability is CDC SVI as published, exposure is the 2020 census, priority is a trade-off a planner sets on a slider, and residents receive context and guidance rather than assessments of their own parcel. We then ask what it takes for a vision-language model's output to count as evidence, grading 1,225 labelled samples per model with six open VLMs on one GPU under verbatim published prompts; the results are published as an evaluation and refused as a claim. The audit record preserves three incidents where the harness caught the system violating its own rules, and 363 Python and 67 app tests form a verifiable evaluation environment. We close with what this architecture cannot do, and why we think an agent in public decision-making should be a steward rather than an oracle.

CCS Concepts

• Information systems → Geographic information systems; • Computing methodologies → Artificial intelligence.

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Keywords

GeoAI agents, accountability, disaster resilience, social vulnerability, provenance, policy enforcement, vision-language models, H3

1 Introduction

Large-language-model agents can now operate geographic information systems with little human intervention [4, 7]. In disaster response that capability meets an unforgiving audience: an emergency manager acting on a tile-level damage estimate, or a resident reading a sentence about their own street, cannot audit a transformer. The prevailing answer is a disclaimer, which locates accountability in the reader. RAY locates it in the system. Following Yang and Zou [6], we treat an agent's claim as valid only when the *outcome* passes executable spatial checks, the *process* is preserved and inspectable, and the claim is *institutionally* authorized — the right actor asserting the right thing at the right resolution — and we add a fourth question raised by third-party data: what can a *reader* do to check this, and may the evidence be kept at all?

This paper describes the deployed system¹ along the four axes of the Track A rubric. *Autonomy* (§2): the pipelines write, run and check spatial analysis on their own, and the places where a human sets a parameter are deliberate. *Robustness and generalizability* (§3): missing data, failing endpoints and mismatched geographies are declared rather than smoothed, by one harness that produced three cases in two states and two hazard types. *Social good alignment* (§4): vulnerability, exposure and priority are defined operationally, the trade-off is the user's, and equity is enforced as rules about who may be told what. *Innovation and reflection* (§5–6): model output treated as governed evidence, three self-caught violations, and the limits of the whole approach.

2 Autonomy Under Constraint

What runs without a human. The data plane is a set of GitHub Actions pipelines. Four keyless federal connectors (USGS earthquakes, NWS alerts, NHC tropical cyclones, NIFC/WFIGS wildfires) refresh hourly and write append-only snapshots and a national watch product. Deep-case builders ingest raw registries and execute the spatial logic end to end: 18,428 CAL FIRE DINS structure points binned to a 265-cell H3 r9 grid with per-severity counts;

¹Live site, app, and repository: <https://rayford295.github.io/ray-resilience/>

CDC SVI tract ranks assigned to cells by point-in-polygon; 2020 census blocks allocated to tiles by centroid; 2,244 reliability-gated cross-view street samples matched to a coverage grid; and, in §5, six vision-language models grading 1,225 labelled images per model through one endpoint contract. No low-level cleaning is done by hand. A hashed dataset registry (134,272 files, ~33 GB) carries a label crosswalk onto one canonical severity scale; CDC's -999 sentinel becomes a counted missing value, never a number; and when the Hugging Face repackaging of the Milton pair set disagreed with the Figshare pair table on 182 of 2,556 pairs, the builder took the table as truth and *recorded* the disagreement instead of reconciling it.

Debugging as a first-class record. Autonomous code fails. The harness's answer is not a retry loop but fail-closed stages whose failures are written down: an over-strict tract-join assertion during the Eaton SVI build was rejected, logged, and corrected in a later run, and the log keeps both rows. Every artifact carries a SHA-256 digest and an audit row (agent, inputs, UTC time); every grid cell must carry an uncertainty block or the build fails.

Where the human is, on purpose. The language-model agent, Ray, answers questions; it does not choose which pipelines run. We declined an LLM task planner deliberately: a model deciding which analyses execute cannot be reconciled with a policy that decides what may be asserted. The human-in-the-loop points are explicit. A user selects a role (resident or planner), which changes what the agent may say. A planner sets the one value-laden parameter the system computes with — an inspection priority $t \cdot \text{damage} + (1 - t) \cdot \text{SVI}$ on a live slider, tiles scored with partial data flagged, every adjustment audit-logged — and can draw an area and ask about it, receiving an answer that covers only the evaluated tiles inside it and highlights the tiles it cited. A resident enters an address and receives one of three answers: covered, outside the evaluated areas, or *not determined* when a layer could not be read. The system will not claim an address is outside areas it failed to look at.

3 Robustness and Generalizability

The Steward Harness sits between every pipeline stage, and between the model and every sentence it emits. *Outcome validity*: CRS assertions, join integrity, sanity bounds, mandatory per-tile uncertainty. *Process validity*: SHA-256 lineage in an append-only audit the app renders from any layer back to source snapshots; the gateway's audit stores a point as its H3 cell and a question as a digest, so a caller can prove their question produced a row without the log retaining what a resident typed about their own home. *Institutional validity*: one YAML policy with two planes of ordered, first-match rules, default deny, validated at load. The *claim plane* scopes what may be asserted by role, evidence tier, resolution and geography; the *distribution plane* scopes what a build may *publish*, and CI verifies the assembled site against it and fails the deploy on violation. One *published_events* list feeds both planes. *Citation by default*: every factual sentence cites an artifact identifier or is refused; the exemption set is closed, so an unanticipated sentence form costs a refusal, not an uncited claim. *Verifiability*: each artifact carries retained > re-derivable > cited-only with weakest-link aggregation and a license attribute; where map-platform terms forbid retention, the harness attests to the request plus a

response digest, a record publishable because it holds no content, and a property test proves no payload reaches the audit.

Real-world complications, declared. *Missing data*: 42 of 295 Palisades images lack EXIF GPS, so the grid covers photographed structures and says so; a source that fails leaves the watch badge reporting *mapped-of-total* and naming the source; a class with $n=30$ is declared as lacking power. *Endpoints*: an ArcGIS error envelope fails the wildfire connector closed instead of reporting zero fires; live adapters time out at 30–60 s and record the timeout; a model call that times out is a recorded LLMUnavailable, and a model the serving stack cannot load is a declared failure, not a zero. *Spatial misalignment*: tract-level SVI on an H3 cell is a downscaling approximation stated per feature; a census block straddling tiles is wholly assigned to its centroid's tile, and no areal interpolation is invented; population outside the evaluated tiles is reported as a total, never mapped into cells the event has no evidence for; post-event Milton imagery is season-cumulative (Debby, Helene, Milton) and every feature carries that attribution caveat.

Generalizability. The same harness, policy and app produced the Eaton Fire 2025 (CA, wildfire), Hurricane Milton 2024 (FL) and Hurricane Ian 2022 (FL) cases; adding an event is a dossier directory and one entry in *published_events*, and the policy contains no geography. The Tier-1 watch is nationwide by construction. Where evidence does not generalize the system says so: Ian's 4,121 further street-view positions ship as a density-only layer because no verifiable per-point severity link exists. The test suite is the robustness evidence: 363 Python and 67 app tests in CI, including a cell-by-cell policy-matrix sweep (every role $\times$ tier $\times$ resolution), adversarial claim checks, connector tests against real error envelopes, and an allowlist drift check that fails on any hand edit. Judges can run it in two commands from the README.

4 Social Good Alignment

Definitions, operationalized. *Vulnerability* is the CDC/ATSDR SVI 2022 percentile rank (overall and four themes) of the census tract a cell falls in, published as a tract quantity and downscaled with the approximation declared; the system adds no index of its own. *Exposure* is 2020 decennial population by block, declared as pre-event population, not who was present. *Damage* is an official or human label — CAL FIRE DINS classes, a reparability label derived from them, or a human perception label on a street-view pair — never a model's guess (§5). *Priority* is $t \cdot \text{damage} + (1 - t) \cdot \text{SVI}$ with t chosen by the planner; the system computes the trade-off, it does not own it. Each definition is a committed artifact with a hashed source snapshot, so a reader who disagrees with a definition can see exactly what to disagree with.

Equity as enforced rules. Fairness here is not a metric but a set of institutional constraints about who may be told what, each a named rule. *No parcel for anyone*: no role receives a parcel-level claim; tile evidence never becomes a statement about a house. *No damage assessment for residents*: they receive exposure context and guidance, not raw damage assessments of their own street, because a false "not significantly affected" costs the reader more than a refusal. *Planner damage at Tier 3 only*: planners see tile-level damage where Tier-3 cross-view evidence exists. Declared unknowns render as prominently as findings; *not determined* is a distinct answer from

Table 1: Six open VLMs and a 7B reference, one pass at temperature 0, quantised weights, same seeded samples and prompt digests. Accuracy / NCSE; columns are not comparable with each other. The closed-model row is RAPID’s published number (same Palisades images; a 150-pair Milton subset).

model	Palisades (5 cl.)	Milton pairs (3 cl.)	Eaton (3 cl.)
qwen2.5vl:7b (ref.)	0.475 / 0.213	0.597 / 0.238	0.910 / 0.049
qwen3-vl:32b [†]	0.554 / 0.176	0.381 / 0.431	0.901 / 0.053
qwen2.5vl:32b ^{†‡}	0.507 / 0.206	0.560 / 0.240	0.931 / 0.039
gemma3:27b	0.529 / 0.187	0.500 / 0.257	0.914 / 0.049
mistral-small3.2:24b	0.505 / 0.203	0.487 / 0.258	0.935 / 0.037
gemma3:12b	0.546 / 0.178	0.427 / 0.287	0.906 / 0.051
qwen3-vl:8b	0.514 / 0.195	0.462 / 0.324	0.924 / 0.041
closed (RAPID)	0.573 GPT-5-mini	0.591 GPT-5.1	—

[†]8k context so the weights fit the card; [‡]95% in VRAM, recorded, because a spilled model is a different run. A seventh model (llama3.2-vision:11b) could not be loaded by the serving stack and is declared a serving failure, not scored.

outside; and no rule authorizes facility claims, so answers about hospitals and shelters default-deny rather than guess.

Community perspectives, honestly. The Milton truth labels are human damage perception, not an instrument reading, and the resident mode speaks in plain language with every qualifier attached. Beyond that we claim little: no community co-designed these definitions, and no local knowledge enters the pipeline yet. What the architecture offers is the *place* where negotiated definitions would go — a declarative, human-readable policy file in which a community’s answer to “who may be told what” is a rule, not a prompt — and the record that would let a community check whether the system kept its word.

5 Model Output as Governed Evidence

The obvious next step for an imagery-rich system is to let a vision-language model (VLM) grade damage. The harness turns that step into a question: under what conditions may a model’s guess become evidence? Model output is a distinct class, `model_derived`, that carries its own measured accuracy or carries nothing; no claim-plane rule admits it; the gateway’s evidence store refuses to load it. We ported the Damage Recognition prompts of RAPID [5] verbatim (Prompt C single image, five DINS classes; Prompt B pre/post pair, three human-perception classes), froze each as a `prompt_sha256` in every record, and served open weights with Ollama on one RTX 3090 (24 GB), so the run is keyless and repeatable. Three cases: Palisades 2025 (RAPID Dataset C2, 295 images, all graded); Milton 2024 (100 pairs per class, seed 2026); Eaton 2025 (300 per class, the $n=30$ minority class fully included; the post-event image only, declared; five classes collapsed onto the set’s three repairability classes). Every prediction is a record with image sha256, truth, prediction, response digest, latency and model digest; an unparseable answer is recorded as such, never coerced. A stage aborts if more than 20% of answers are off-schema, a metric leaves $[0, 1]$, fewer than 95% of graded samples fall inside the event’s committed grid, or any product lacks an uncertainty block. We report accuracy and $\text{RAPID’s NCSE} = \frac{|y - \hat{y}|}{(K-1)}$.

What the runs say. Table 1: (i) on a single post-event image every open model beats the 7B reference and the best is within two points of the paper’s closed models, with no monotone size effect — gemma3:12b (0.546) is within a point of qwen3-vl:32b (0.554); (ii) on pre/post pairs no open model beats the 7B reference, and each fails with a different fixed bias — qwen3-vl:32b calls 258 of 300 pairs Severe (Mild recall 0.01), gemma3:12b calls 262 Moderate, mistral 213 — so a vote across models would not be a fix; (iii) Eaton’s $n=30$ class is weak for every model (recall 0.03–0.43) while the large classes sit above 0.9; (iv) the qwen3 family spends its time in hidden reasoning before a 60-character answer, and on Milton 24 of qwen3-vl:8b’s pairs hit the client timeout, recorded as unanswered (8.3%, under the bound).

What the harness did with it. the agreement grids sit beside the human-labelled grids they are evaluated against, flagged `model_derived`; the publication planner ships them as an evaluation; the claim plane has no rule that admits them; the gateway skips them and records the exclusion. A reader can verify every number from the records without a GPU. What no one can do — and what the system therefore does not do — is turn a 0.93 into a sentence about a street.

6 Reflection

Three times the harness caught its own system. (1) A parcel-level DINS file reached the public site through an asset-copying script, violating four written statements while breaking zero policy rules, because the claim plane governs what the agent *asserts* and nothing had asserted the file. The fix was a new plane — distribution — with a planner and a CI gate; the public surface fell from 30 files to 16, and the incident report is committed beside the code it indicts. (2) The citation check originally fired only on sentences with digits, so “*your neighborhood was not significantly affected*” passed uncited; inverted to citation-by-default, acceptance on a draft set fell from 9/13 to 7/13 sentences, the two refusals being exactly the uncited reassurances a resident should not receive. (3) The gateway’s evidence store read every event on disk, ignoring `published_events` and `model_derived`; the Palisades VLM run would have been cited as tile facts. Both planes now publish from one list, the store records every exclusion and refuses to start over “everything on disk”. The lesson repeats: a constraint written in one place and enforced in another is enforced nowhere.

Limitations. The builders read a ~33 GB corpus on one workstation, so third parties can verify the committed artifacts and audit logs but not regenerate them. The VLM numbers are one pass on 4-bit weights on a 24 GB card; Milton and Eaton are seeded samples, and the closed-model Milton reference used a different subset. The gateway runs locally without a hosted endpoint, auth or rate limiting, so the public demo ships without chat; the re-derivable regime has run only against a stub. Three deep cases are not national analysis capability, and the app says so.

Ethics and the societal role of the agent. The disclosure risk in this domain is concrete: per-structure damage is a statement about a family’s home, and our first incident shows how easily it leaks without a rule that names it. We therefore treat the resolution cap, the role denials and the content-free audit not as features but as the conditions under which an agent may speak to the public at

all. The value judgements — which index counts as vulnerability, how damage trades against it — are handed to people, in a file they can read and a slider they can move, and the system records that they did. Autonomous-GIS agents [4, 7] and operational platforms such as TDIS [3] show what such systems can compute; registries (EM-DAT [1], GDACS [2]) show what they can record. What none carries, per row, is what resolution of claim the evidence authorizes and whether a reader may keep it. Our position is that an agent in public decision-making should be a steward of evidence rather than an oracle: it should compute more than it is allowed to say, say only what it can cite, and leave a record — including of its own failures — that the public it serves can check.

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Claude (Anthropic) to polish the language of this manuscript; the system, analyses, and claims were designed, written, and verified by the authors, who take full responsibility for the content.

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